



Starting *Small* with AI: Building Trustworthy KPI Dashboards

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Not a BI tool. A governance-first AI validation layer for institutional KPI pipelines — free, open-source, and built for lean IR offices.

THE 10-STEP VALIDATION FRAMEWORK

I · DEFINE Describe the KPI precisely

01	Register Metadata	All required fields present and non-empty
02	Validate Definition	Formula operator · Time period · Definition precision
03	Entity Coherence	Cross-system join risk: Low / Moderate / High

II · VALIDATE Test for quality and governance

04	Data Quality	Outliers ($z \geq 2.5\sigma$) · Step changes ($\geq 25\%$) · Flags only, never auto-corrects
05	AI Use Assessment	Documents permitted techniques · Excludes black-box scoring
06	Governance Check	Owner · Steward · Limitations · Equity risks · Signoff HARD GATE
07	Readiness Decision	Ready for Dashboard Use · or · Not Ready PUBLISH GATE
08	Summary Report	JSON audit trail · CSV executive summary · PNG time-series chart

III · ENHANCE Advisory: design quality and standards

09	Design Quality	SMART (5 criteria) + Tufte visualization principles (5 criteria) ADVISORY
10	Industry Standards	EDUCAUSE · AGB · NACUBO alignment · Score: X/9 ADVISORY

"Dashboard quality cannot exceed KPI governance quality. The 10-step framework is the engine. AI is the power steering."

THREE-MODULE PIPELINE

kpi_parser.py

Claude API

Plain-language intake
· Human review before validation
→ `parsed_kpi.json`

kpi_readiness_10step.py

Pure Python

10-step engine · No API key required
→ `_report.json`
→ `_summary.csv`
→ `_chart.png`

kpi_improver.py

Claude API

Refinements · HTML dashboard · LLM survey analysis
→ `_dashboard.html`
→ `_suggestions.txt`

STEP 9 — DUAL ASSESSMENT FRAMEWORK

SMART

Doran, 1981

Metric construction — is the KPI specific, measurable, achievable, relevant, and time-bound?

Tufte / Few
2001 · 2006

Metric interpretation — will it chart clearly, honestly, and with sufficient comparison context?

Governance
Steps 1–6

Institutional accountability — ownership documented, human signoff required before publication.

GOVERNANCE AT EVERY TOUCHPOINT

- ✓ Human review before validation
- ✓ Incomplete governance blocks publication
- ✓ Equity risks required in every report
- ✓ Anomalies flagged — never auto-corrected
- ✓ All AI output labeled as draft for review
- ✓ LLM-agnostic — swap any API provider

NORCO COLLEGE PILOT — 13 KPIS REVIEWED

- 10/13 KPIS lacked a comparison benchmark or target
- 2 KPIS showed significant trend changes with no annotation or explanation
- All 13 were displayed on separate pages, obscuring cross-metric relationships

Conducted with Hayley Ashby, Ed.D., Norco College (RCCD).
Presented at REACH Research Analytics Summit, Newport RI, April 2026.



Scan for code, README
& supplementary materials

github.com/ElmYgl/kpi_readiness_10step

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Presented at AIR Forum 2026, Washington D.C.

Proposal #2234062

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SIDE 1 OF 2

Get Started in Minutes

Three entry levels · Any IR office · No vendor dependency required for core validation

github.com/ElmYgl/kpi_readiness_10step

Full paper: AIR Forum 2026 · Proposal #2234062
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INSTALLATION & SETUP

Level 1 Core Validation Only

Requires: Python 3.8+ only · No API key · No additional install

```
# Download from GitHub, then:
python kpi_readiness_10step.py
```

Runs demo immediately. Generates `_report.json` and `_summary.csv`. Edit the `KPIInput` block in `main()` to validate your own KPI.

Level 2 With Plain-Language Parser

Requires: Python 3.8+ · Claude API key · `pip install anthropic`

```
pip install anthropic
# Windows PowerShell — set API key:
[System.Environment]::SetEnvironmentVariable(
    'ANTHROPIC_API_KEY', 'sk-...', 'User')
# Mac / Linux:
export ANTHROPIC_API_KEY='sk-...'
python kpi_parser.py
```

Paste any KPI description in plain language. Three data options: include in description, upload Excel/CSV, or no data yet. Human review required before validation runs.

Level 3 Full Pipeline

Requires: Python 3.8+ · Claude API key · All three modules

```
pip install anthropic openpyxl matplotlib
python kpi_parser.py
# After validation completes:
python kpi_improver.py
```

`kpi_improver.py` auto-finds the latest `_report.json` and generates the board-ready HTML dashboard and AI-suggested refinements.

EVERY PIPELINE RUN PRODUCES	
<code>_report.json</code>	Full 10-step audit trail
<code>_summary.csv</code>	Executive summary · Opens in Excel
<code>_chart.png</code>	Time-series with anomalies marked
<code>_dashboard.html</code>	Board-ready · Opens in any browser
<code>_suggestions.txt</code>	AI refinement candidates for review
White = <code>kpi_readiness_10step.py</code> (pure Python) · Gold = <code>kpi_improver.py</code> (Claude API)	

AI GOVERNANCE — STEP 05

✓ PERMITTED	✗ EXCLUDED
✓ Entity resolution across systems	✗ Automated individual decisions
✓ Anomaly detection on aggregates	✗ Black-box scoring
✓ NLP for survey comments	✗ Auto-correction without review
✓ Synthetic data — clearly labeled	✗ Ungoverned AI output published

SESSION LEARNING OUTCOMES

- L01** Identify which AI techniques are appropriate for KPI validation — and which must be excluded — with documented rationale
- L02** Apply the three-entry-level pipeline to begin validating KPIs within your own office context
- L03** Implement a reusable 10-step validation framework aligned with EDUCAUSE, AGB, and NACUBO standards

DEMO RUN — FIRST-YEAR RETENTION RATE

Step 07 READY ✓ Dashboard Use **Step 09.1** 5/5 SMART — all criteria met
Step 09.2 4/5 Tufte — add benchmark **Step 10** 8/9 Strong Alignment

Tufte (Step 9.2) and AGB (Step 10) flag the same gap: missing comparison benchmark. Convergent finding — two independent frameworks, one recommendation.

CITE THIS WORK

Yglesias, E. (2026). Starting small with AI: Building trustworthy KPI dashboards. Paper presented at *AIR Forum 2026*, Washington, D.C.
github.com/ElmYgl/kpi_readiness_10step

EDUCAUSE Data Governance Framework 2023	AGB Dashboard Metrics for Boards 2022	NACUBO KPI Framework 2024
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